

Inv-SHAF

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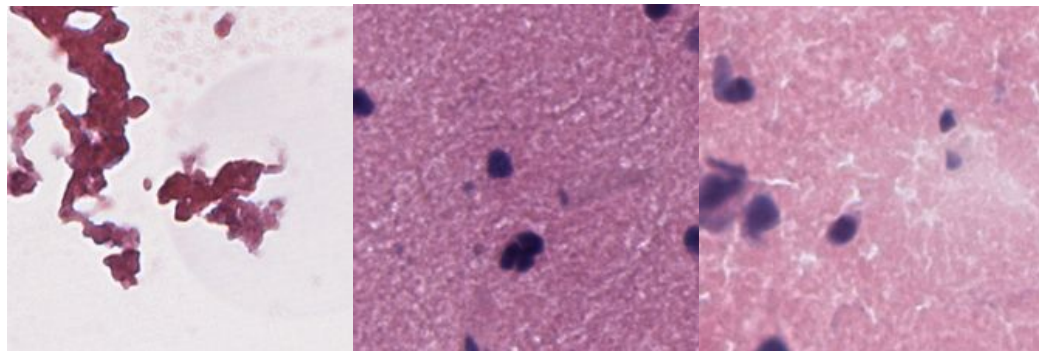
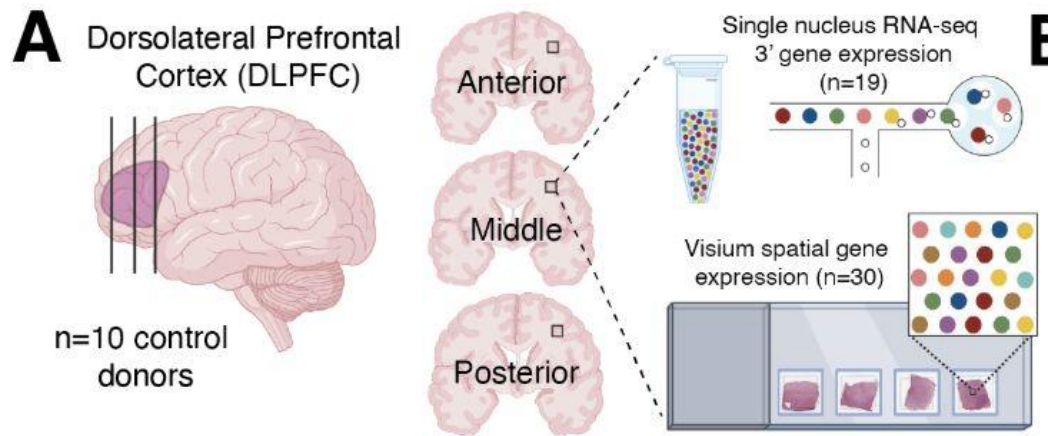
Motivation

- **Spatial transcriptomics is powerful but expensive**: measuring gene expression at every spot is costly and low-throughput.
- **Histology is cheap and ubiquitous**: every tissue section already has morphology, so we want to predict gene expression from image structure.
- **The core scientific goal**: learn a mapping from tissue appearance to spatial gene expression that generalizes across donors, not just within one experiment.
- **Why active learning here**: if labels are expensive, we should query the spots that most improve robustness instead of labeling everything.

Invariance, Batch Effects, and Why We Use Them

- **Batch effects** are donor- or experiment-specific shifts: staining, tissue handling, sequencing artifacts, and morphology differences that are not the true biological signal we want.
- **Invariance** means learning features that predict genes while being **less dependent on donor identity**.
- In our project, donor is the environment. A good model should capture relationships that stay stable across those environments.

Dataset Overview



- **Dataset:** Spatial DLPFC Visium
- **Environments / donors:** 8 anterior DLPFC donor sections
- **Split used in active learning:**
 - test set: 15%
 - initial labeled seed: 1%
 - unlabeled pool: 84%
- **Scale:**
 - 30,868 spots
 - 100 selected genes
 - 224 x 224 RGB patch per spot

Acquisition Functions

- We start with a small labeled seed and repeatedly query the unlabeled pool.
- Each round acquires about **0.25% of the dataset** up to **10%**.
- Scoring Functions:
 - **Random**: donor-stratified random sampling
 - **Uncertainty**: query the highest MC-dropout variance
 - **Invariance**: query the largest disagreement between Model 1 and the donor-specific cheater

$$V_{\text{Inv}}(x) = \text{mean}(\hat{y}_{\text{inv}}(x) - \hat{y}_{\text{donor}}(x))^2$$

- **Spatial Min**: prioritize patches with the lowest Moran's-I-weighted predicted expression

$$I = \frac{N}{W} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$$

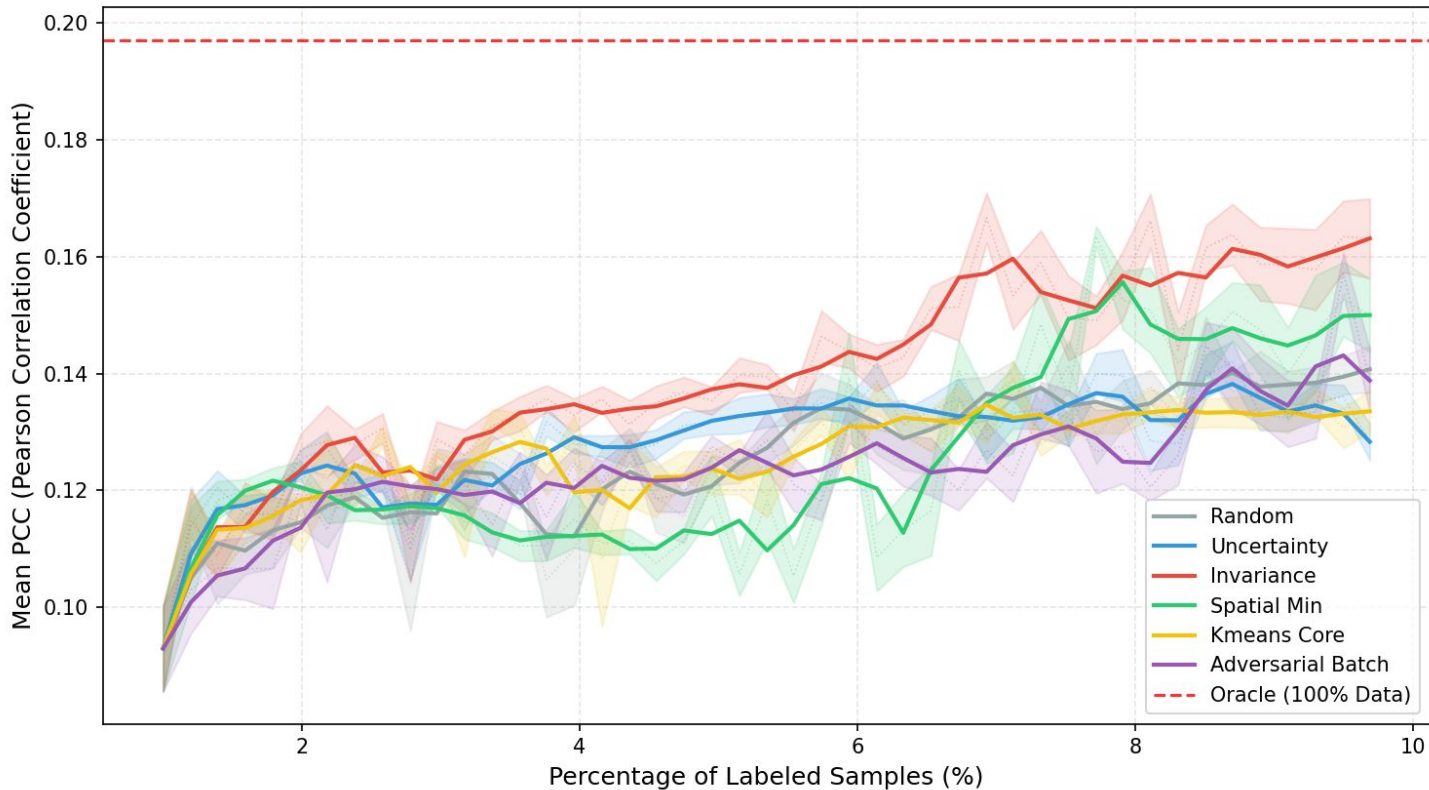
- **KMeans Core**: cluster feature embeddings and query points closest to cluster centroids.
- **Adversarial Batch**: query patches with the lowest donor-discriminator (the most donor-specific samples)

Architecture

- **Backbone**: UNI2-h foundation model encodes each histology patch into a 1536-D embedding
- **Model 1: invariant learner**
 - Bottleneck: 1536 -> 256
 - **Head A**: gene predictor for 100 genes
 - **Head C**: discriminator, donor classifier used to remove batch-specific information
 - **Intuition**: Model 1 learns a shared tissue representation that predicts gene expression while being discouraged from encoding donor-specific batch effects
- **Model 2: donor specific cheater**
 - one linear gene-prediction head per donor
 - trained directly on the raw UNI embedding

Result

Spatial Gene Expression Prediction: Active Learning Comparison



Takeaway and Future Directions

- We have come up with a way to **leverage mechanistic invariance** across donor environments to make active learning more label-efficient.
- This turns batch effects into a **useful acquisition signal**: donor-specific disagreement helps identify the labels that are most informative for learning robust biology.
- **Environments are everywhere**: in this project the environment is the donor, but many datasets also contain slide effects, lab effects, staining differences, sequencing chemistry shifts, and site-to-site variation.
- The same acquisition principle could be extended to any **batch or environment label**, not just donor identity